Thomas C. Fraser

Curriculum Vitae
February 27, 2023

Personal

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Address	Perimeter Institute for Theoretical Physics
	31 Caroline St. North, Waterloo, Ontario N2L 2Y5 Canada
Citizenship	Canadian
Date of Birth	27/11/1994

Education

2018 – Present	Ph.D., Physics *Perimeter Institute For Theoretical Physics, Waterloo, ON*
2017 – 2018	M.Sc., Theoretical Physics, Quantum Information Thesis: A Combinatorial Solution to Causal Compatibility *Perimeter Scholars International Program, Perimeter Institute For Theoretical Physics, Waterloo, ON*
2012 – 2017	B.Sc., Mathematical Physics, Astrophysics Specialization *University of Waterloo, Waterloo, ON*

Papers

2022 T. C. Fraser
"A sufficient family of necessary inequalities for the compatibility of quantum marginals"
arXiv preprint arXiv:2211.00685

2020 D. Schmid, T. C. Fraser, R. Kunjwal, A. B. Sainz, E. Wolfe, R. W. Spekkens
"Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory"
arXiv preprint arXiv:2004.09194

2020 T. C. Fraser
"A Combinatorial Solution to Causal Compatibility"
Journal of Causal Inference 8(1), 22-53

2018 J. R. Rinehart, J. H. Béjanin, T. C. Fraser, M. Mariantoni
"Machine Learning Peeling and Loss Modelling of Time-Domain Reflectometry"
arXiv preprint arXiv:1804.04756

2018 T. C. Fraser, E. Wolfe
"Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario"
Physical Review A, 98(2), 022113

Talks

2022 The IQC-QuICS Math and Computer Science Seminar
A sufficient family of necessary inequalities for the quantum marginals problem

2021 Beyond IID in Information Theory 9
Simultaneous Spectral Estimation and the Quantum Marginal Problem

2021 PI Grad Student Conference, Waterloo, Canada
Simultaneous Spectral Estimation and the Quantum Marginal Problem

2019 IQST Seminar at University of Calgary, Canada
Sudoku on steroids - The combinatorics of causal inference

2016 Quantum Networks Conference at International Institute for Physics, Natal, Brazil
Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario

Teaching

Winter 2021 Quantum Information - Perimeter Scholars International (PSI) Program
Teaching Assistant

Winter 2021 Quantum-classical gaps in causal structures - PSI Winter School
Student project co-lead

Fall 2018 Physics 1 - Phys 111 - University of Waterloo
Teaching Assistant

Awards & Scholarships

2022 *Natural Sciences and Engineering Research Councilof Canada (NSERC)*
Three years, Grant No. 411301803

2018 *UW Graduate Award & Marie Curie Award*
Sufficiently High Entrance Average, Department Selected

2018 *The Joanne Cuthbertson and Charlie Fischer Graduate Student Award*
Support for an exceptional Masters student from the 2017/2018 PSI class

2017 *Governor General's Silver Academic Medallion*
Highest Graduating Average in Graduating Class

2016 *Mike Lazaridis Scholarship*
Theoretical Physics Fellowship at Perimeter Institute

2015 *Xerox Research Centre of Canada Limited Award*
Best work-term report "Acoustic Modelling Using Mel-Frequency Cepstral Coefficients"

2015 *C. C. Lim Physics Prize*
Top Marks in Undergraduate Thermodynamics

2013 *Don E. Brodie Scholarship*
Highest Experimental Physics Lab Performance

2012 *A. Donald Maynes Scholarship*
Outstanding Academic Record

2012 *BMO Undergraduate Entrance Scholarship*
Outstanding Academic Average

Professional Experience

September 2015 – January 2016	*Research & Development Data Scientist* Sysomos, Toronto, ON, Canada
September 2014 – May 2015	*Video Game Developer* Lunarch Studios, Waterloo, ON, Canada
January 2014 – May 2014	*Mathematics Tutor* Humber College, Toronto, ON, Canada
June 2011 – August 2011	*Solar Panel Technician* OVG Solar, Inc, Renfew, ON, Canada

Programming

Languages C, C++, Python, Mathematica, Matlab, Actionscript, JavaScript

References

Rob Spekkens
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